SAM VOLUME I

# Home: information on an endless chain of two-dimensional closed spheres

Chapter XIII

Sean Brady
Originator and conceptual director
Research development with OpenAI ChatGPT and Codex
Edition of 25 October 2026 (text of the unified expanded edition, 19 October 2026)

**Abstract.** A Home is a closed two-dimensional ledger with an interior side and a parent-domain exterior side. What an exterior observer calls bulk volume is read as ordered information on that surface, and the surface's own completion can become the floor of a new interior, so that the construction continues as an endless chain of closed spheres, each the exterior of the next. This chapter carries the idea as an idea: the custody recurrence $E_{n+1} = E_n + I_n - O_n$ with its nonnegative exterior account, the conditional radius floor, the distinction between gross output and net growth, surface response, quantum custody and what an exterior observer can learn, the re-indexing of a completed boundary into a new floor, the formation-transfer model and its counterexample to an unsupported inference, the entropy ledger for receipts, and a finite program for deciding the fate proposal. The text is that of Sections 9–19 of the submitted Chapter XII, unchanged, renumbered as a chapter of its own.

**The idea, stated once.** Every earlier chapter of this volume treats a boundary as a place where accounting changes hands: a source's retained account against its field, a clock's reading against the road it has travelled, a horizon's accessible information against the inventory behind it. The Home proposal takes that habit to its limit. It says that the natural object of the theory is not a volume filled with matter but a closed two-dimensional surface carrying ordered information, with an interior that is itself such a surface's exterior, without end in either direction. Equal intervals and filled geometry are then the exterior observer's conventions, not what the ledger is made of. Nothing in this chapter proves that. What it does is make the proposal precise enough to be wrong: a recurrence whose exterior account is provably nonnegative, a radius floor that follows under a stated closure, a custody rule that forbids free copying, a transfer model that shows what a change of inventory label cannot do by itself, and a finite program of checks with stopping rules. The chapter ends with that program because the proposal's standing is what it has yet to pass.

**Status, 25 October 2026 (Part 2).** The Home is register target T-12 and the crossing postulate T-14 (*Targets register*, 23 October 2026); neither has been built natively, and the evidence appendix's conditional reading stands. Three results of the record touch the idea without testing it, and are named here as connections, not as support. Volume II's Li-6 receiver algebra, reproduced natively from the archive's exact operands (H002783, 25 October 2026), found the component a receiver discards to be a finite phase register read one plane per Write, carrying a conserved label and a gauge the receiver sees only through part of its response: a closed surface account in the matter language, with nothing of the gate's hidden stage in it. The Volume II gate run (H002782) found retained information monotone and log-convex but not completely monotone, which is the shape a surface ledger with a floor would have. And the Riemann window run (H002784) found the program's prime structure scale-free, a function of ratios and invisible on the equal-interval ruler, which is the same claim about conventions this chapter makes about volume. The chapter's own next step is unchanged from its last section: one surface model shared with Chapters VII and XI that reproduces the custody recurrence under a source-selected action, with the numerical ledger frozen while it is tested.

## 1. A completed Home: retained proposal and conditional closure

The second arc of the original chapter describes a Home as a closed two-dimensional ledger with an interior side and a parent-domain exterior side. Apparent bulk volume is interpreted as ordered information on the surface, rather than an additional ontic three-dimensional interior. Tagged child histories retain their custody under recursive maps. These are organizing postulates of the Home proposal, not deductions from the Schwarzschild metric.

The original lifecycle has three stages. A pre-completion collapse state has no filled point interior; localization remains assigned to the forming surface. During formation, the origin-matter lane and the captured outside-origin baryon lane establish a formation inventory. At completion, the source coordinate reaches $A = 1$ and the ledger is sealed under the proposed crossing rule. The historical identifiers H000791 and H000794–H000797 record this construction.

Chapter V already limits how strongly the last step may be stated. Its source coordinate satisfies $A(r_s) = 1$; this is not the condition $A_0 + A = 1$. Its finite-impulse and finite-distance results depend on the source law. In the relevant infinite-approach route, a calculated source satisfying the radial null-energy condition is screened by the theorem; otherwise the actual energy-condition violation or flux must be propagated. Chapter XII cannot turn this conditional result into an unconditional physical law of infinite crossing cost.

Accordingly, the completed Home account below is conditional on a source-selected completion dynamics and boundary

law consistent with Chapter V. It may impose an empty post-completion crossing relation as part of that model, but must then derive the stress, work and causal response that realize it. Coordinate-time delay in a Schwarzschild exterior does not by itself establish infinite local crossing cost. The physical statement requires an operational trajectory and energy account.

## 2. Exterior balance and the radius floor

Once the conditional completion is stipulated, define $M_{\mathrm{form}} > 0$ as the frozen formation inventory and $E_n \geq 0$ as a mutable exterior mass-equivalent inventory. Let $I_n, O_n \geq 0$ be realized incoming and outgoing mass-equivalent receipts over step $n$. The proposed recurrence is

$$\begin{gathered} E_{n+1} = E_n + I_n - O_n, \qquad 0 \leq O_n \leq E_n + I_n, \\ M_n = M_{\mathrm{form}} + E_n. \end{gathered} \tag{1}$$

The explicit upper bound on $O_n$ is what preserves nonnegativity. If $E_0 \geq 0$, induction gives $E_n \geq 0$ for every subsequent step. Summing the recurrence gives the exact cumulative identity

$$E_N = E_0 + \sum_{n=0}^{N-1} I_n - \sum_{n=0}^{N-1} O_n, \qquad \sum O_n \leq E_0 + \sum I_n. \tag{2}$$

These are ledger theorems. They do not specify a physical emission rate or show that every allowed sequence occurs.

If the Schwarzschild mass–radius adapter is adopted at each quasi-static step,

$$R_n = \frac{2G(M_{\mathrm{form}} + E_n)}{c^2}, \qquad R_{\mathrm{form}} = \frac{2GM_{\mathrm{form}}}{c^2}, \tag{3}$$

then $R_n \geq R_{\mathrm{form}}$. An area-based record count with fixed area per unit satisfies

$$\frac{N_n}{N_{\mathrm{form}}} = \left(\frac{R_n}{R_{\mathrm{form}}}\right)^2 = \left(1 + \frac{E_n}{M_{\mathrm{form}}}\right)^2 \geq 1. \tag{4}$$

This consequence requires both the adapter and an unchanged microscopic area unit. If the completion mechanism alters either, the area-memory formula must be rederived.

A worked example clarifies gross output versus net change. In arbitrary mass units choose $M_{\mathrm{form}} = 10$ and $E_0 = 2$. An interval with $I_0 = 3$ and $O_0 = 1$ ends at $E_1 = 4$: the Home emits while its total mass grows from 12 to 14. Next take $I_1 = 0$ and $O_1 = 3$, obtaining $E_2 = 1$ and total mass 11. Finally $I_2 = 0$, $O_2 = 1$ exhausts the exterior account and leaves the floor 10. The cumulative output is 5, paid by initial exterior inventory 2 plus inflow 3. At no step is the formation floor debited.

A positive floor has a sharp thermodynamic consequence. If all incoming supply ceases and exterior inventory drains, the proposed output must eventually cease or change its law. A temperature formula by itself cannot authorize continued emission from a frozen reservoir. The boundary transition strengths and available states must implement the floor. This is where Chapter VII's thermal account must meet the lifecycle rather than remain a separate analogy.

## 3. Continuous evolution, vacuum exteriors and surface response

A continuum approximation to the recurrence is

$$\dot{E} = \dot{M}_{\mathrm{in}} - \dot{M}_{\mathrm{out}}, \qquad E \geq 0. \tag{5}$$

At $E = 0$, the boundary rule must prevent $\dot{M}_{\mathrm{out}} > \dot{M}_{\mathrm{in}}$. A formal ordinary differential equation that continues to negative $E$ violates the model's own domain. One can impose a constrained flow, but a microscopic theory should explain its origin through allowed transitions and detailed energy payments.

There is also a geometry requirement. A spherically symmetric vacuum exterior in general relativity has a constant Schwarzschild mass parameter locally; time-varying mass caused by actual inflow or radiation requires nonvacuum stress in the relevant region [XII.4]. Therefore (3) is at most an instantaneous or quasi-static adapter during exchange. It is not a complete spacetime solution for arbitrary receipt sequences. Matching an emitting surface to its exterior requires the energy flux and junction conditions.

The same surface cannot be assigned independently chosen optical, thermal and gravitational-wave properties. A hard reflective surface, a dissipative membrane and an absorbing horizon can share an approximate exterior mass profile while having different responses. Chapter XI's impedance and Chapter VII's thermal transitions must be compatible with the Home custody rules. A boundary that stores a pulse must say where its energy goes and how a later receipt releases it. A boundary that permanently hides information must specify the channel seen by each side.

These obligations also prevent a purely semantic resolution of the singularity question. Calling the boundary an information surface does not calculate its stress or establish a nonsingular evolution. Conversely, a successful surface model would not need a classical bulk interpretation to be meaningful if it provided a consistent state space, dynamics, observation map and tested exterior limit. Those are the appropriate requirements for the proposal on its own terms.

## 4. Quantum custody and what an exterior observer can learn

The original Home rules describe unresolved horizon matter as unavailable to either side. A failed incoming attempt retains its exterior receipt; a retained post-completion receipt has a finite sequence of attempted inward operations followed by a realized outward operation. The terminal outward receipt debits scalar parent-visible mass. This is a proposed event grammar, not yet a Hamiltonian or a quantum instrument.

A physical version can represent an allowed step by an instrument $\{\mathscr{E}_r\}$ indexed by receipts $r$. Each map is completely positive and trace nonincreasing; their sum is trace preserving when all outcomes are included. Probabilities are $p_r = \mathrm{Tr}\,\mathscr{E}_r(\rho)$, and conditioned states are $\mathscr{E}_r(\rho)/p_r$. Conservation then requires the mean receipt energies and changes in source, surface and control energy to agree. Naming an outcome "OUT" does not supply this equality.

The assertion that each receipt eventually exits also needs a precise quantifier. Every member of a finite set may have a finite exit schedule, yet an indefinitely replenished population need not empty at any finite time. A stochastic model can have eventual exit with probability one while having an infinite mean waiting time. Neither a uniform finite completion time nor a finite luminosity follows from individual eventual realization. The waiting-time distribution is part of the observable dynamics.

The parent-facing scalar quotient severely limits accessible information. If two internal states $\rho$ and $\sigma$ induce the same output channel state, no measurement on that output distinguishes them. More generally, a completely positive trace-preserving channel $\Phi$ contracts trace distance,

$$\frac{1}{2}\|\Phi(\rho)-\Phi(\sigma)\|_1 \le \frac{1}{2}\|\rho-\sigma\|_1. \tag{6}$$

This is the operational frame already used in Chapter VI. A child history that never changes any allowed parent output is not parent-observable. A proposed test must identify which output distribution, response function or conserved scalar actually depends on the child data.

Preserving custody is also different from duplicating an arbitrary quantum state. Side tags and ancestry labels may record classical provenance, but a coherent state cannot be copied freely into independently readable ledgers. The Home map must specify whether it is an isometry, a coarse-graining channel, a conditional instrument or some other well-defined transformation, and carry correlations with the environment accordingly.

## 5. Re-indexing a completed boundary into a new floor

The original handoff identity is

$$A_0 = \frac{A_H}{4\pi D}, \tag{7}$$

where $A_H$ is the unit completion value in the relevant normalization and $D$ is the spatial-dimension count. To avoid confusion with physical distances elsewhere in this chapter, write $D_{\mathrm{count}} = 3$ here. With $A_H = 1$ and $R = 4D_{\mathrm{count}} = 12$, the identity yields $A_0 = 1/(12\pi)$ and $RA_0 = 1/\pi$. The independence of the product from $D_{\mathrm{count}}$ holds only when the relation $R = 4D_{\mathrm{count}}$ is retained.

The arithmetic is clear; the direction and mechanism of the handoff need more definition. The working chapter alternates between a child floor and a floor of the parent accumulation. A physical construction should label the input domain $\mathscr{H}_P$, the boundary system $\mathscr{H}_B$ and the output domain $\mathscr{H}_C$ explicitly. "Re-indexing" might mean a change of description of one state, a new effective subsystem, or actual creation of a causally distinct domain. Those meanings have different conservation and observational requirements.

At minimum, a candidate transition must specify the prepared boundary state, the map acting on it, the conserved quantities, the energy source for its realization and the matching conditions that fix the new inventory. The factor $4\pi D_{\mathrm{count}}$ supplies none of those by itself. A unit closure divided by an angular and dimensional factor is a normalization proposal, not a transition amplitude or a nucleation probability.

If the new domain is causally inaccessible, its internal cosmological history cannot be checked directly from the parent. The strongest available test would be a relation among accessible parent quantities, such as mass, area, response and receipt statistics, fixed by the same map that assigns the child floor. Without such a relation, the continuation remains a possible interpretation of the bookkeeping rather than an empirically distinct cosmological model.

## 6. Formation history and the primordial-inventory interface

The formation ledger in the original proposal separates origin matter, placed in a primordial-hole or trapped-accumulation lane, from outside-origin baryons captured during formation. Its frozen mass is

$$M_{\mathrm{form}} = M_{\mathrm{origin}} + M_{B,\mathrm{form}}. \tag{8}$$

This is a useful decomposition only when the formation process determines both terms and their distributions. It does not yet produce the mass function used in Chapter X, the baryon abundance used in Chapter VIII, or the thermal initial conditions used in Chapter IX.

A mass window alone is not a formation history. A physical primordial-hole scenario must specify the time or temperature of formation, the relation of object mass to the available horizon or source scale, the initial abundance, subsequent accretion, merging and evaporation, and correlations with the surrounding material. The closing review supplies conditional temperature estimates for particular mass windows. Those depend on a radiation-era horizon-mass adapter and a formation efficiency; they should not be promoted to native SAM outputs without those inputs.

The appropriate cross-chapter test is conservation through the whole chain. Formation receipts establish a distribution; that distribution evolves through the thermal history; the surviving population enters the later inventory and halo model. Energy returned during formation or evaporation must appear in the thermal and radiation budgets. If a radius floor leaves stable remnants, their abundance and mass density must be included. A free choice at each stage would remove the predictive force of the proposed common origin.

## 7. A solved formation-transfer model

### 7.1 Changing custody does not automatically create pressure

A homogeneous toy model can distinguish a change of inventory label from a change of cosmic dynamics. Let $\rho_u$ be the mass-equivalent density of unbound nonrelativistic matter and $\rho_H$ that of a population of completed Homes, also treated as nonrelativistic objects on large scales. This approximation deliberately ignores individual strong-field regions and asks only

how a declared transfer affects a coarse-grained background. Use $N = \log a$ and a prime for $d/dN$.

Suppose a fraction $\lambda\, dN$ of the unbound inventory transfers during each logarithmic expansion interval, with $\lambda \geq 0$ constant. If transfer is lossless and both populations have zero effective pressure, the continuity equations are

$$\rho_u' + 3\rho_u = -\lambda\rho_u, \qquad \rho_H' + 3\rho_H = \lambda\rho_u. \tag{9}$$

For initial values at $a = 1$, direct integration gives

$$\rho_u(a) = \rho_{u0} a^{-3-\lambda}, \tag{10}$$

$$\rho_H(a) = a^{-3}\left[\rho_{H0} + \rho_{u0}(1 - a^{-\lambda})\right]. \tag{11}$$

Adding the equations or solutions yields $\rho_u + \rho_H = (\rho_{u0} + \rho_{H0})a^{-3}$. Transfer changes custody but leaves the total dust dilution unchanged. In particular, it does not create a constant-density term or negative pressure simply because the destination is a completed surface.

This is not a derivation of the actual Home formation rate. It is a counterexample to an unsupported inference: a conserved conversion of one pressureless category into another cannot by itself explain cosmic acceleration. A mechanism producing a different effective pressure must supply its stress, work or exchange law. If binding energy is released, the radiation or other receiving sector must appear explicitly instead of being silently removed from the homogeneous account.

### 7.2 Including a paid radiative fraction

Let a fraction $0 \leq \beta \leq 1$ of the transferred mass-equivalent energy enter radiation and the remainder enter Homes. Then

$$\rho_u' + 3\rho_u = -\lambda\rho_u, \tag{12}$$

$$\rho_H' + 3\rho_H = (1-\beta)\lambda\rho_u, \tag{13}$$

$$\rho_r' + 4\rho_r = \beta\lambda\rho_u. \tag{14}$$

The first solution remains (10); the second becomes $a^{-3}[\rho_{H0} + (1-\beta)\rho_{u0}(1-a^{-\lambda})]$. For $\lambda \neq 1$, the radiation solution is

$$\rho_r(a) = a^{-4}\left[\rho_{r0} + \frac{\beta\lambda\rho_{u0}}{1-\lambda}(a^{1-\lambda} - 1)\right]. \tag{15}$$

At $\lambda = 1$, the bracket is $\rho_{r0} + \beta\rho_{u0}\log a$, the continuous limit of the same expression. All densities remain nonnegative for $a \geq 1$ and nonnegative initial values. For $\lambda > 1$, both $a^{1-\lambda} - 1$ and $1 - \lambda$ are negative, so their ratio is still nonnegative.

Summing the three equations gives

$$\rho_{\rm tot}' + 3\left(\rho_{\rm tot} + \frac{p_{\rm tot}}{c^2}\right) = 0, \qquad p_{\rm tot} = \frac{1}{3}\rho_r c^2. \tag{16}$$

The exchange terms cancel, while radiation retains its additional expansion work. Consequently the mass-equivalent energy in a comoving volume is not constant when radiation is present; it decreases by the pressure work. That does not represent an unpaid local transfer. It is exactly the homogeneous continuity balance of the adopted geometry.

For an elementary numerical check choose $\rho_{u0} = 1$, $\rho_{H0} = \rho_{r0} = 0$, $\lambda = 1/2$, $\beta = 1/4$ and $a = 4$. The densities are $1/128$, $3/512$ and $1/1024$, respectively. Their sum is $15/1024$, smaller than the lossless dust value $1/64 = 16/1024$ because the produced radiation has performed expansion work. The example uses arbitrary density units and does not assign those parameters to SAM. It supplies an exact test for any implementation claiming to conserve the same transfer ledger.

## 8. The same present inventory can have different futures

A second solved example shows why a present-day complement cannot determine fate. Adopt a spatially flat homogeneous comparator with pressureless matter and a vacuum-like component of pressure $p_X = -\rho_X c^2$. Permit an explicitly prescribed transfer $Q = 3\xi H\rho_X$ from that component into matter, where $0 < \xi < 1$ is constant. The continuity equations become

$$\rho_X' = -3\xi\rho_X, \qquad \rho_m' + 3\rho_m = 3\xi\rho_X. \tag{17}$$

Their exact solutions are

$$\rho_X(a) = \rho_{X0} a^{-3\xi}, \tag{18}$$

$$\rho_m(a) = \rho_{m0} a^{-3} + \frac{\xi\rho_{X0}}{1-\xi}(a^{-3\xi} - a^{-3}). \tag{19}$$

The right-hand term is nonnegative for $a \geq 1$. At $\xi = 0$ the noninteracting cosmological-constant solution is recovered; at $\xi = 1$ the matter solution acquires the limit $3\rho_{X0}a^{-3}\log a$ in addition to its initial dust term.

The initial fractions can be chosen to equal the same $1/\pi$ and $(\pi-1)/\pi$ used in the original adapter. Nevertheless, for every fixed $0 < \xi < 1$, both densities vanish in the infinite-expansion limit. At late times their ratio approaches $\rho_m/\rho_X = \xi/(1-\xi)$ and the total density is asymptotic to $\rho_{X0}a^{-3\xi}/(1-\xi)$. The effective equation of state approaches $w_{\rm eff} = -(1-\xi)$.

Using the flat Friedmann relation then gives $H \propto a^{-3\xi/2}$. Integrating $\dot{a} = aH$ yields the asymptotic power law

$$a(t) \propto t^{2/(3\xi)}, \qquad H(t) \sim \frac{2}{3\xi t}, \tag{20}$$

up to a shift of the time origin and a positive normalization. Thus $H$ tends to zero, whereas the noninteracting model with the same initial inventory tends to a positive constant. For $0 < \xi < 2/3$ the power exceeds one and expansion still accelerates. Acceleration and a nonzero limiting Hubble rate are therefore distinct properties.

This toy model is not proposed as a fit or a new SAM law. Its exchange was chosen openly to expose a logical dependency. The value of the present inventory does not fix $Q$, and the future changes when $Q$ changes. A Home-based explanation of the background would have to derive an exchange law, show that it is compatible with local conservation and observations, and state whether its pressure assignment is appropriate. The toy family supplies an exact comparison against which that derivation can be checked.

## 9. Global causal access requires the whole expansion history

A completed local boundary and a cosmological event horizon are different objects. In a spatially flat homogeneous metric, the comoving radial distance that an outgoing light signal can cover after time $t$ is

$$\chi_{\mathrm{future}}(t) = c\int_t^{t_{\mathrm{end}}} \frac{dt'}{a(t')}, \tag{21}$$

where $t_{\mathrm{end}}$ is the future endpoint of the chosen solution. A finite integral defines a finite future comoving reach for that observer. The corresponding proper event-horizon distance, when applicable, is $a(t)\chi_{\mathrm{future}}(t)$. This is a global statement about an entire future solution, not a local statement at one value of the density fraction.

For exponential expansion $a \propto e^{H_\infty t}$ with constant positive $H_\infty$, the proper distance is $c/H_\infty$. For a future power law $a \propto t^p$ extending to infinite time, the integral converges only when $p > 1$. In that case the proper horizon distance is $ct/(p-1)$, which grows without bound even though the comoving future reach is finite. If $p \le 1$, the integral diverges and this future event horizon is absent.

Applied to (20), the criterion is $\xi < 2/3$. The interacting example can therefore have an event horizon while its Hubble rate tends to zero. Neither the existence of a horizon nor its proper size follows from multiplying a road fraction by a background number. A future recollapse, finite-time endpoint, changing spatial curvature or different propagation law would require a new causal calculation.

This distinction also clarifies the Home continuation. Calling a domain inaccessible to its parent may be a rule of the proposed quantum channel, a feature of the global geometry, or both. A geometric horizon does not automatically impose a particular quantum custody map. A custody map that removes all parent-readable distinctions does not automatically determine a spacetime causal diagram. The model should identify which statement it makes and derive the relation if it claims both.

## 10. An explicit entropy ledger for receipts

Energy conservation does not determine how much information a receipt contains. A minimal classical example has two possible receipt values with probabilities $p$ and $1-p$. Its dimensionless Shannon entropy is

$$H_2(p) = -p\log p - (1-p)\log(1-p), \tag{22}$$

with the continuous endpoint convention $0\log 0 = 0$. A deterministic outward receipt can carry nonzero energy and zero receipt entropy. A random receipt can carry entropy even when the mean energy is held fixed by suitable channel energies. Thus the scalar mass quotient and the information revealed by its realization are separate observables.

If a classical register $R$ stores the receipt and the conditioned boundary state is $\rho_r$, the joint state is $\sum_r p_r |r\rangle\langle r| \otimes \rho_r$. Its dimensionless von Neumann entropy is

$$s(BR) = H(p) + \sum_r p_r s(\rho_r). \tag{23}$$

Tracing out the register gives $\overline{\rho}_B = \sum_r p_r \rho_r$, and the mutual information is $I(B:R) = s(\overline{\rho}_B) - \sum_r p_r s(\rho_r)$. Adding the marginal entropies without subtracting this mutual information double-counts correlations. Any claim that the surface preserves or releases information must therefore specify whether it means a joint entropy, a conditioned entropy or an accessible record.

For a quasistatic thermal comparator with Hawking temperature $T_H$, removal of energy $\delta E > 0$ gives a horizon entropy change approximately $-\delta E/T_H$. An ideal equilibrium massless radiation gas with zero chemical potential satisfies $p = u/3$ and $Ts = u + p$, hence has entropy-to-energy ratio $4/(3T)$, suggesting a positive combined balance when $T = T_H$. That example is not a proof for a filtered, nonthermal or correlated finite pulse. The actual channel state and greybody propagation determine its entropy, and finite reservoir corrections belong to Chapter VII's exact transition account.

A formation floor makes the bookkeeping more restrictive. Once the mutable exterior energy vanishes, a thermal law cannot continue spending the frozen inventory unless the floor rule is changed. If an external controller resets receipt registers, its work and entropy export also belong in the enlarged account. A complete quantum instrument can preserve global information while making the parent output highly mixed, but that is a statement about the full transformation, not a consequence of the word "custody." The receipt model supplies the variables needed to formulate such a transformation without assuming its answer.

The solved examples identify three different senses of conservation. A finite Home recurrence conserves its receipt account after all inflows and outputs are included. A homogeneous cosmology conserves stress through the continuity equation, including pressure work in an expanding volume. A quantum channel conserves normalization while its accessible entropy can change because correlations move between subsystems. None of these statements can replace the others. A parent mass quotient does not establish a cosmological equation of state, and a conserved global quantum state does not make every detail readable by the parent. Conversely, failure to recover an internal history from one scalar channel does not prove that the full evolution destroys information. A complete continuation mechanism should provide a translation among these three accounts and state the approximation under which each is valid. This requirement is constructive: the receipt sequence, homogeneous transfer solutions and channel-state formulas above give explicit limiting cases against which such a mechanism can be tested.

## 11. A finite program for deciding the fate proposal

The next calculation should begin with an observation map rather than an asymptote. Define a road observable on finite routes, state its clocks and energy conservation law, and derive its effect on luminosity distance, angular size, radial distance and redshift. Equations the distance-duality ratio of Chapter XII, Section 8 and the inferred-rate relation of Chapter XII, Section 8 then provide immediate consistency checks. A model that only

changes the name of a standard parameter will not separate itself from the adapter.

The next dynamical step is to derive an evolution law from the retained source ensemble. It must establish whether $1/\pi$ is an attractor, an initial condition or merely a normalization. It must also identify the pressure and exchange terms that determine $H(a)$. A finite-redshift prediction with an uncertainty and failure criterion is more informative than an unreachable future limit.

For the Home continuation, the shortest route is a single surface model shared with Chapters VII and XI. It should reproduce the proposed custody recurrence, respect the conditional closure results of Chapter V, and yield both a thermal response and a perturbative response. The absence of one observable does not make the model untestable if another allowed output carries information, but that output must be calculated before a comparison is selected.

Finally, the numerical ledger should remain frozen during each test. If a road factor is changed to preserve distance duality, an inventory is shifted to fit a new cosmological likelihood, or a boundary impedance is adjusted to match ringdown, the change should be recorded as a new model branch. This practice allows the count program to learn from failure without retrospectively labeling every close expression a prediction.